

Mitigating Production Risks

Strategic Decoupling of Critical Components

SCM Consulting Team, Group 7

Abhirup Saha | Aman Kumar Sinha | Pallabi Keot

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Prepared for: Chief Operating Officer, Maruti Suzuki India Ltd.

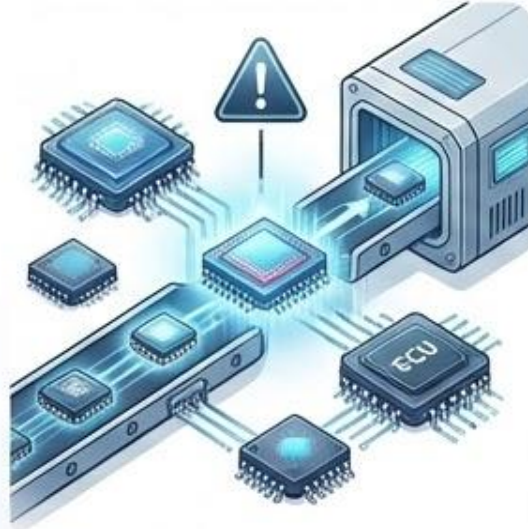


THE JIT VULNERABILITY & PROBLEM STATEMENT



The Efficiency Trade-off

Maruti Suzuki's highly efficient Just-In-Time (JIT) model eliminates safety stock to save costs, creating a severe vulnerability to global supply shocks.



Single-Source Dependency

The supply chain relies heavily on single sources for critical electronic components, such as Electronic Control Units (ECUs).



The Bullwhip Effect

Without strategic decoupling buffers, minor upstream delays in semiconductor fabrication instantly halt massive downstream assembly lines.



Business Impact

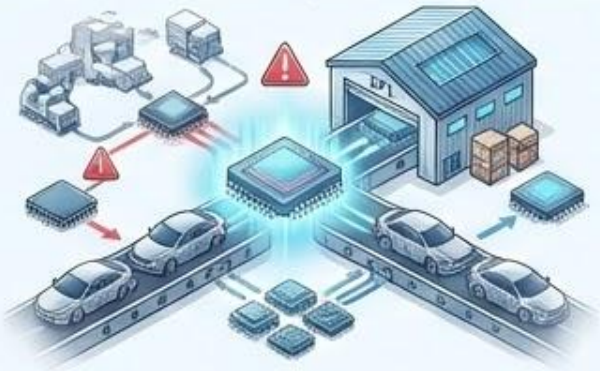
According to the FY 2022-23 Annual Report, the company suffered a production loss of 170,000 vehicles entirely due to these shortages.

Evaluating Solution Options

To resolve the efficiency vs. resilience trade-off, we evaluated two primary strategies before arriving at an immediate, demand-driven solution.

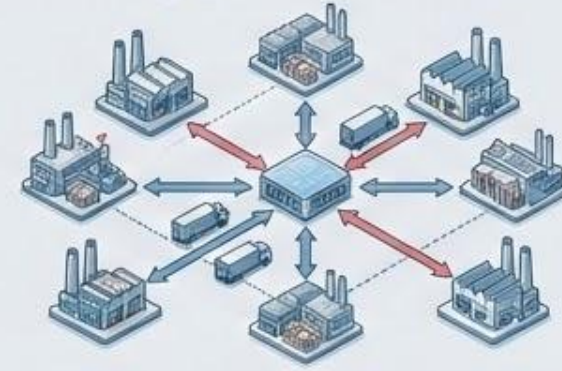
RECOMMENDED

Option A: Strategic Buffering



Concept: Demand-Driven MRP (DD-MRP) | Move critical, high-risk ECUs away from the rigid JIT model. Establish a strategic physical buffer in a nearby 3PL warehouse to instantly decouple the assembly line from upstream supplier delays and guarantee production

Option B: Supplier Diversification



Concept: Near-Shoring & Multi-Sourcing | Develop new chip suppliers in alternate regions like India or Vietnam. While theoretically sound for the long term, this requires years for technical qualification and incurs massive, immediate switching costs.



1. Classify Risk Components



2. Calculate DD-MRP Buffers



3. Setup 3PL Decoupling Point



4. Automate ERP Orders

Exhibit 1: DD-MRP Buffer Sizing for ECUs

System Assumptions:



Average Daily Usage (ADU): 5,000 units



Average Lead Time (LT): 30 days



Lead Time Factor (LTF): 0.5



Variability Factor (VF): 0.5



Order Cycle: 10 days



Yellow Zone

Strategic Purpose:
Expected usage during lead time

Calculation Formula:
 $ADU \times LT$



Total Units:
150,000



Red Base

Strategic Purpose: Protection against variability (Demand & Supply)

Calculation Formula:
 $A_{\text{Yellow}} \times LTF$



Total Units:
75,000



Red Safety

Strategic Purpose: Extra protection for critical, long lead-time components

Calculation Formula:
 $D_{\text{RedBase}} \times VF$



Total Units:
37,500



Green Zone

Strategic Purpose: Order generation trigger & frequency controller
Calculation Formula: $A_{\text{RedBase}} + \text{RedSafety}$



Total Units:
50,000



Total Buffer Sizing

Total Units:
312,500



Exhibit 2: Financial Justification

Lost Profit (Cost of Stockout)

₹1,360 Cr

170,000 lost cars × ₹80,000 profit margin per car

Annual Holding Cost (Resilience)

₹6.25 Cr

312,500 max chips × ₹200 holding cost per year

Business Conclusion

The pure lean supply chain model failed to account for the massive difference in value between the component and the finished good. The cost to hold a maximum safety buffer of chips for an entire year (₹6.25 Cr) is **less than 0.5%** of the profit lost from the stockout (₹1,360 Cr).



Questions?

Thank you for your attention.

